LESSON PLAN

AIR WEAPON (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Geometrical characteristics of aerofoil, definition of terms NACA series, various forces e.g. lift, drag and moments, theories of Lift production center of pressure, aero dynamic center, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Geometrical characteristics of aerofoil, definition of terms NACA series, various forces e.g. lift, drag and moments, theories of Lift production center of pressure, aero dynamic center, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back.
5. Specific Objectives	10.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer and various concepts of Aerodynamics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of NACA series models; Safety: Handle models carefully; Supervision: Instructor monitors model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aerofoil?
(b) Define the boundary layer in aerodynamics.
(c) List the four main forces acting on an aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Geometrical characteristics of aerofoil • Definition of terms NACA series • Various forces lift drag and moments	L & Ds	PPT, IB	15 min
(b)	• Theories of Lift production • Center of pressure and aerodynamic center • Force and pitching moment concepts	L & Ds	PPT, PDF	20 min
(c)	• Coefficient of lift definition • Variation with aspect ratio • Effect of sweep back on lift	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between center of pressure and aerodynamic center.
(b) How does increasing the aspect ratio affect the coefficient of lift?
(c) Describe the role of sweep back in lift generation.

12. Home Assignment and Follow Up	Prepare a brief note on how the NACA series classification system defines aerofoil camber and thickness for use in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min